LINDEF: Lightweight Real-Time Network Intrusion Detection & Containment

Introduction

Cybercrime: Illegal activity performed by individuals using computers and the Internet (U.S. Immigration and Customs Enforcement)

- **600%** increase in cybercrime during Covid-19 Pandemic (Lallie et. al)
- Cyber Attackers have leveraged **Generative AI** to automate and personalize cyber attacks, posing a higher threat

What is Network Intrusion?

- **Network Intrusion:** Unauthorized access to a digital network, usually with the aim to steal sensitive data or disrupt operations
- There has been an increase in the number of entry points for hackers to exploit with the transition to remove work and cloud computing

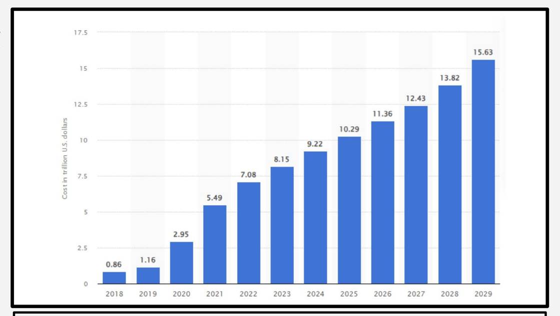


Figure 1. Estimated Cost of Cybercrime Worldwide 2018-2029. Petrosyan, A. (2026, January 16). *Global cybercrime estimated cost 2025*. Statista. https://www.statista.com/forecasts/1280009/cost-cybercrime-worldwide/

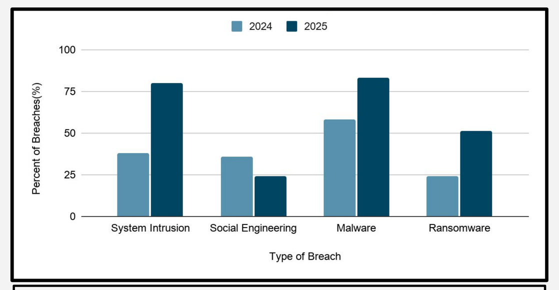


Figure 2. Type of Cybersecurity Attack vs Percent of Breaches Accounted For. Verizon. (2025, April 23). Verizon's 2025 Data Breach Investigations Report: System intrusions behind 80% of APAC breaches. https://www.verizon.com/about/news/2025-data-breach-investigations-report-apac

Review of Literature

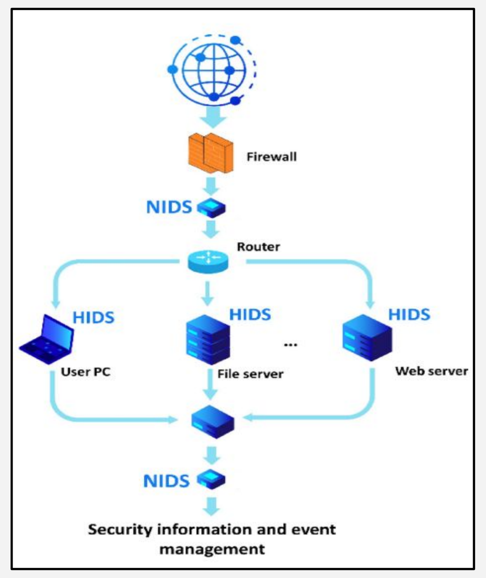


Figure 3. NIDS Role in Attack Prevention *Hozouri, A., Mirzaei, A., & Effatparvar, M. (2025). A comprehensive survey on intrusion detection systems with advances in machine learning, deep learning and emerging cybersecurity challenges. Discover Artificial Intelligence, 5(314).* https://doi.org/10.1007/s44163-025-00578-1

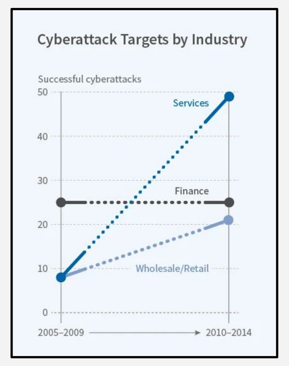


Figure 4. Kamiya, S., Kang, J.-K., Kim, J., Milidonis, A., & Stulz, R. M. (2018). *What is the impact of successful cyberattacks on target firms?* (Working Paper No. 24409). National Bureau of Economic Research. https://doi.org/10.3386/w24409

After a company suffers a data breach:

- Average loss of 1.1% in Market Value
- 3.2% percentage drop on year to year sales growth rate
- Average Immediate Stock Loss of 1.12%, estimated to be $607 million

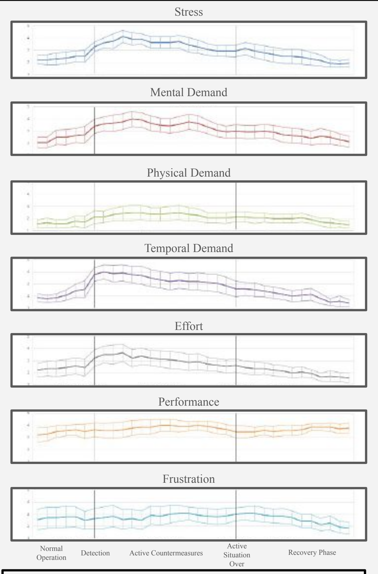


Figure 4. *Average Emotional Journey During an Attack* Virtanen, T. (2024). The psychological effects of continuity threatening cyber incidents. In M. Lehto (Ed.), Proceedings of the 23rd European Conference on Cyber Warfare and Security (Vol. 23, No. 1, pp. 601–608). Academic Conferences and Publishing International. https://doi.org/10.34190/eccws.23.1.2268

Current Solutions

Rule-Based Intrusion Detection System(Aickelin, 2007)	Anomaly-Based Intrusion Detection System (Chandola, 2009)
• Manually written detection tools • Requires constant rule updates	• Detects deviations from normal network traffic • Extensive training data required and high false-positive rate
Cloud-Based Endpoint Detection System (Viharika, 2025)	**Signature-Based Intrusion Detection System (Guide, 2024)**
• Monitors activities on devices using a network • Privacy concerns and alert fatigue	• Checks network traffic against known patterns • Requires frequent updates

Implication

- Many small businesses and organizations are unable to invest the time and money into these expensive detections systems
  - Opens up vulnerability for attacks
- In order to prevent the loss of data and money, it is important to find a reliable intrusion detection system
  - Small businesses and organizations can better ensure the safety of their stakeholders while prioritizing efficient resource allocation and utilization.

Goal

- Develop an autonomous network intrusion detection framework
  - Fast, light-weight, easily accessible, dynamic, and accurate
  - Can accurately detect and classify the risk of network intrusion incidents
  - Doesn't affect performance of other activities

Methodology and Procedures

Data

UNSW-NB15

- Modern Network Intrusion Dataset
- 9 Attack Categories + Normal Traffic
- Approximately 2.5 Million Records
- 49 Features per Flow

NSL-KDD

- Benchmark Dataset for IDS Research
- 147K Records
- 4 Attack Categories + Normal Traffic
- 41 Features per Flow

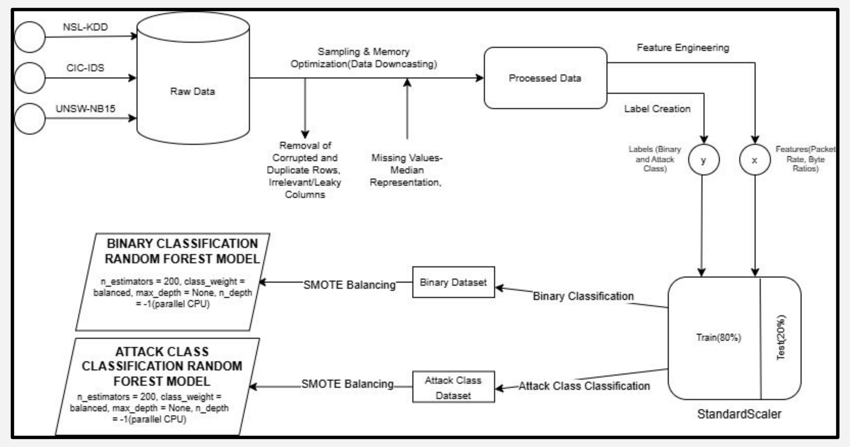
CIC-IDS

- CSV Files with flow-based features
- 80+ Features
- 2.8 Million Records
- 14 Attack Types

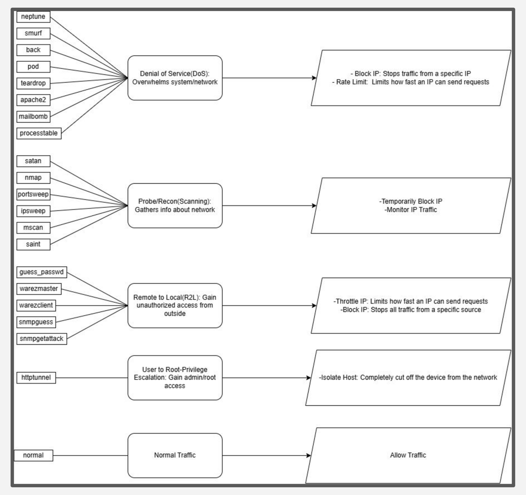
When Combined:

- More comprehensive coverage of attack types
- Better generalization to prevent overfitting
- Larger training data
- Increased feature robustness
- Can be implemented into real, diverse scenarios

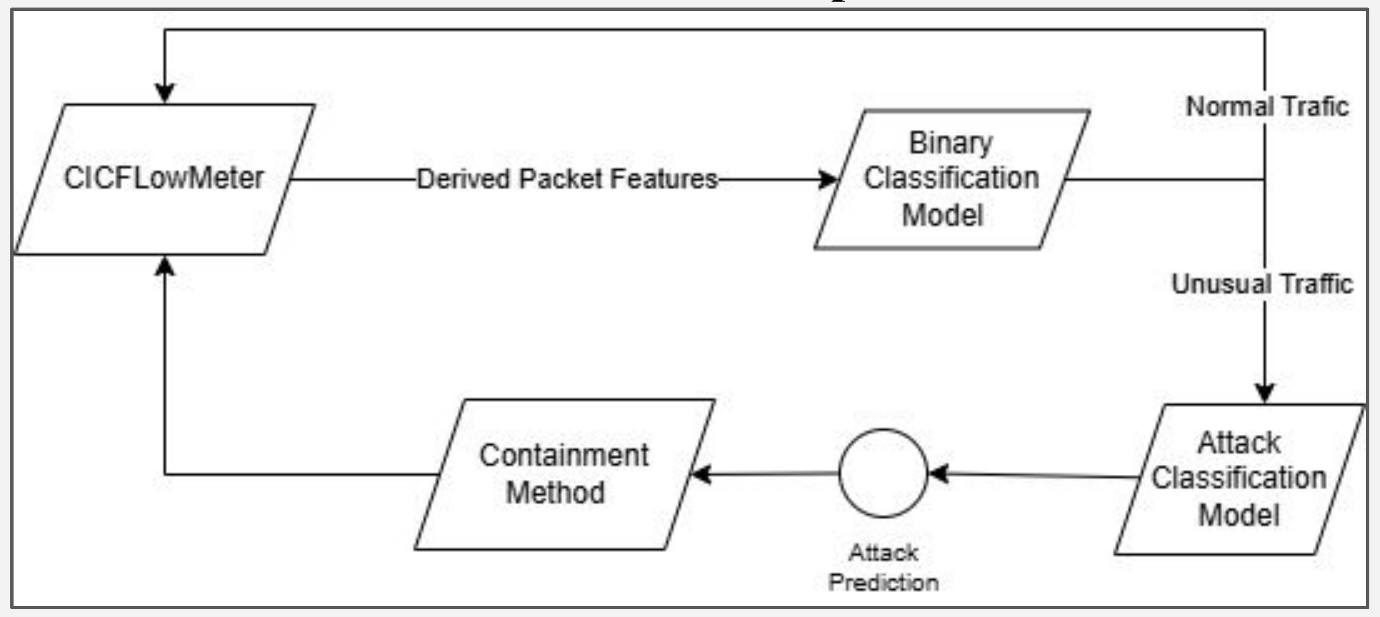
Model Training



Classification Method



Live Detection Pipeline

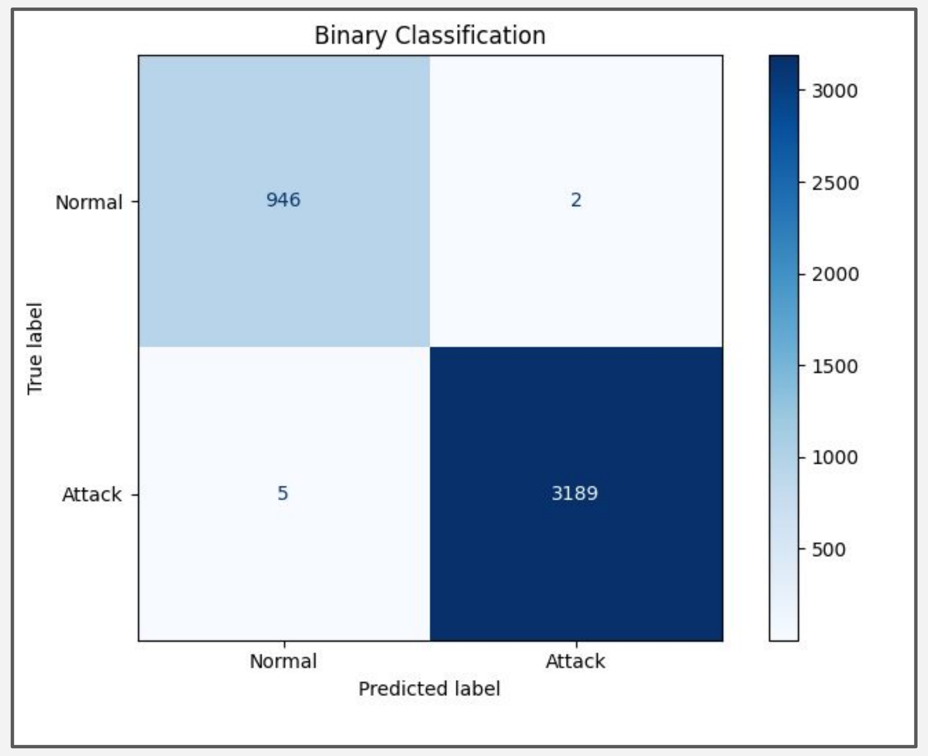


All visuals in Methodology made by student using draw.io.

Results

- Accuracy: 0.9983
- Precision: 0.9983
- Recall: 0.9983
- F1-score: 0.9983

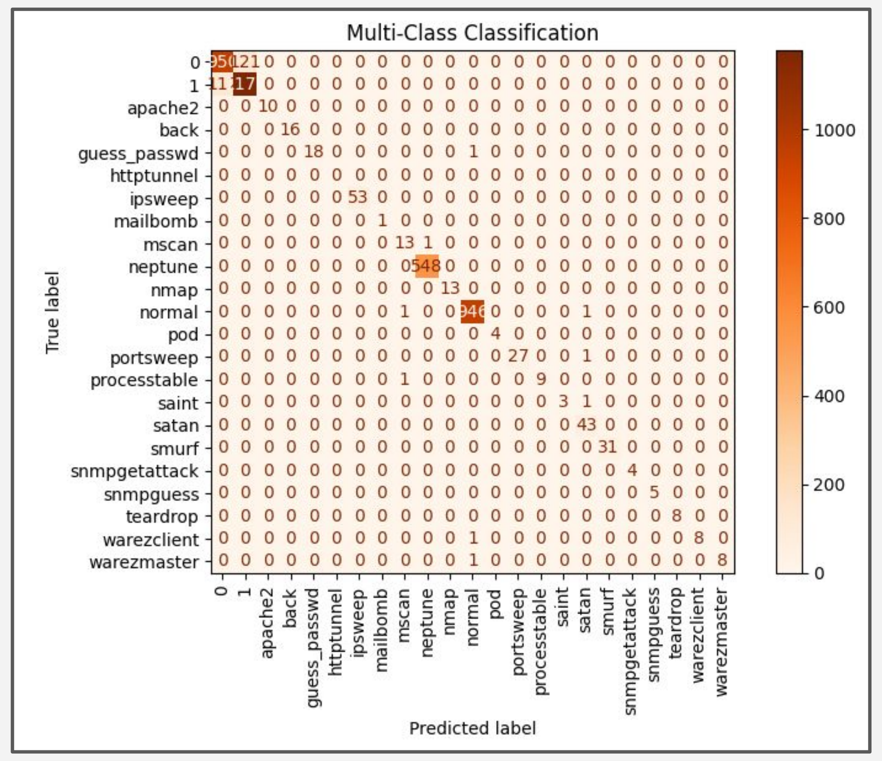
Binary Classification Model



Confusion Matrix made in Python using SciKit-Learn, NumPy, and Matplotlib

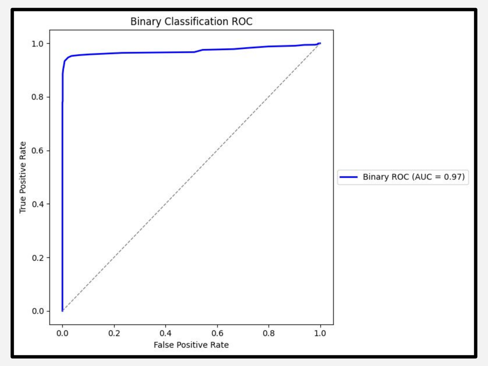
Attack Classification Model

- Accuracy : 0.9415
- Precision: 0.9416
- Recall : 0.9415
- F1-score : 0.9415

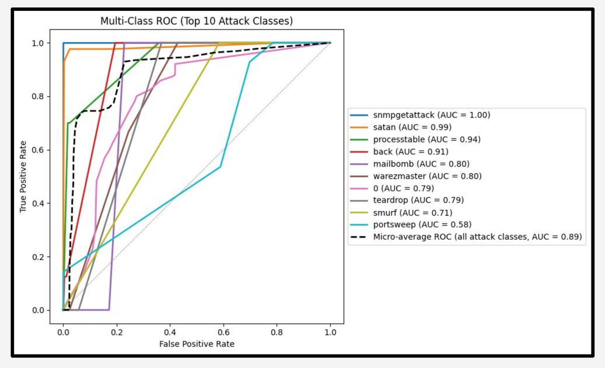


Confusion Matrix made in Python using SciKit-Learn, NumPy, and Matplotlib

ROC-AUC Analysis



Binary Classification Model AUC: 0.97



Attack Classification Total AUC: 0.89

ROC-AUC Curve Graphs were made by students in Python using SciKit-Learn, Matplotlib, Numpy libraries.

Example Attack Logs

Time	Status	Attack	Severity	Confidence	Response
12:01:05	ATTACK	neptune	HIGH	0.987	BLOCK_IP
12:01:06	BENIGN	normal	NONE	0.995	ALLOW
12:01:07	ATTACK	guess_passwd	HIGH	0.912	THROTTLE_IP
12:01:08	ATTACK	httptunnel	CRITICAL	0.978	ISOLATE_HOST
12:01:09	BENIGN	normal	NONE	0.998	ALLOW
12:01:10	ATTACK	smurf	HIGH	0.945	BLOCK_IP
12:01:11	ATTACK	nmap	LOW	0.87	BLOCK_IP
12:01:12	ATTACK	warezmaster	HIGH	0.923	THROTTLE_IP
12:01:13	BENIGN	0	NONE	0.996	ALLOW
12:01:14	ATTACK	apache2	HIGH	0.951	BLOCK_IP

Example Log created in Python using Pandas, NumPy, Time, Joblib, OS, Streamlit, and Excel.

Statistical Analysis

	LINDEF	Signature-Based IDS	Anomaly-Based IDS	Cloud-Based Endpoint Detection	Rule-Based IDS
Detection Accuracy(%)	Binary- 99.83% Attack Class- 94.15%	94%-98%	85% – 95%	96% – 99%	90% – 95%
False Positive Rate(%)	Binary- 0.21% Attack Class- 0.37%	< 1%	3% – 5%	2%	1% – 2%
Average Latency(ms)	Binary- 54.18ms Attack Class-53.79ms	< 5ms	10 – 50 ms	50 – 200 ms	5 – 15 ms
RAM Usage(MB)	Binary - 18.52MB Attack Class- 8.32MB	50-200MB	1 - 4GB	Varies (< 5% of host)	200-500MB
	Statistical Analysis of LINDEF was constructed in Python using SciKit-Learn, NumPy, Pandas, Matplotlib, Joblib, psutil, Time, OS, google.colab.	Statistical Analysis of IDS Methods constructed using: Shah, S. A. R., & Issac, B. (2018). Performance comparison of intrusion detection systems and application of machine learning to Snort system. Future Generation Computer Systems, 80, 157–170. https://doi.org/10.1016/j.future.2017.10.016. Waleed, A., Jamali, A. F., & Masood, A. (2022). Which open-source IDS? Snort, Suricata or Zeek. Computer Networks, 213, 109116. https://doi.org/10.1016/j.comnet.2022.109116. Wirianda, I. B. A., Huwae, R. B., & Jatmika, A. H. (2024). Testing the effectiveness of intrusion detection systems (Snort, Suricata, and Zeek) against SYN flood attacks. Jurnal Bumigora Information Technology, 7(2). https://doi.org/10.30812/bite.v7i2.5226. Zaitsev, E. (2025, December 10). CrowdStrike achieves 100% detection and zero false positives in MITRE ATT&CK Enterprise evaluations. CrowdStrike. https://www.crowdstrike.com/en-us/blog/crowdstrike-achieves-100-percent-mitre-attack-enterprise-evaluation/			

Discussion

- The **results of the LINDEF model** show:
  - Comparable or better accuracy than alternative IDS solutions
  - Comparable or better recall than alternative IDS solutions
  - Slightly worse or comparable average latency than alternate IDS solutions
  - Comparable or better RAM usage and model size than alternate IDS solutions
- Despite having **lower or similar average latency values** compared to alternate IDS containment methods, it showed better values for one or more of the other statistics compared to each alternate IDS solution- **LINDEF has specific advantage over each alternate solutions**
- Resulting effects of the **lower latency values are negligibl**e as systems which will use LINDEF will not be extremely traffic heavy
- ROC-AUC Curves reveal **Attack Classification Model is performing very well, while Binary Classification Model is performing exceptionally**

Conclusion

- The implementation of LINDEF into **real-time environments** will provide a **light-weight** solution to **monitor and prevent network intrusion attacks** with **similar results** to comparative methods **without the cost, maintenance, and resources required**
- Limitations:
  - **Lack of testing** in large enterprise or high-bandwidth networks
  - May struggle with **zero-day attacks** due to lack of previous training to these new attacks
  - Testing of model took place on only one machine, opposed to the **typical hardware environments** these tools are used in
  - Cyber Attackers may **craft traffic** in order to deceive these models
  - **Black-box model**
- Future Research:
  - The researchers will continue to **fine-tune** the model in order to **lower the latency** required by the model.
  - Testing of the LINDEF model in **high-bandwidth and large enterprise networks** in order to **test and improve the functionality** of the model in these scenarios.
  - **Data** that the models were trained on can be **expanded to large enterprise network traffic** in order to better prepare the LINDEF model for **real-time deployment**.

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